

EdgeCount: Statistical Theory and Algorithmic Efficiency

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Abstract

This vignette details the statistical framework underlying the **EdgeCount** package. First, the Random Graph with Given Expected Degrees (RGGED) model is described. Second, the derivation of edge-count statistics is presented. Third, the algebraic optimizations that allow the package to calculate connectedness statistics with linear time complexity are explained. Last, the implementation of vectorized functions contained in the package is discussed, and it is shown that they have optimal time complexity.

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1 Introduction

The **EdgeCount** package provides methods to analyze the connectedness of vertex sets in large, sparse, undirected graphs. Whether analyzing interaction networks or group-element (called term-element) networks, the main question is:

Is an observed number of edges, called edge count, within a set (or between two sets) significantly higher than expected by chance?

To answer this, the package implements a statistical framework built upon the Random Graph with Given Expected Degrees (RGGED) model [7, 8]. The RGGED serves two distinct but related purposes depending on the analysis type:

- Fair comparison (scoring): The RGGED accounts for vertex degrees, which often vary by orders of magnitude. By using analytical expressions for the likelihood of edge counts in this model [18], the package allows for the fair and fast comparison of edge counts across different sets and graph topologies.
- Null model specification:
 - For interaction networks (unipartite graphs), the RGGED is utilized directly as the null model. The statistical significance of a set’s connectedness is determined by comparing the observed edge count to the distribution under the RGGED on that specific set of vertices.
 - For term-element networks (bipartite graphs), the situation is more complex. Because analysis typically involves ranking multiple terms and filtering for existing connections (observed count of at least 1), the RGGED probability acts as a local score or effect size estimator. A second level of modeling—simulating the random selection of element sets—is then required to estimate False Discovery Rates (FDR) and assess global significance.

2 Random graph model

Let $G(V, E)$ be an undirected graph with no loops and no multiple edges.

- Let $N = |V|$ be the order of the graph (number of vertices).
- Let $M = |E|$ be the size of the graph (number of edges).
- Let k_i be the degree of vertex v_i .
- Let $\langle k \rangle$ be the average vertex degree.

Call the degree sequence of G the nonincreasing sequence of all its vertex degrees. An obvious null model for G is that of a random graph whose realizations always preserve the degree sequence. This model is referred to as the Random Graph with Fixed Degree Sequence (RGFDS). However, it is known that the analytical treatment of the RGFDS is a difficult problem [3, 16, 10, 17]. Namely, deriving simple analytical expressions for the likelihoods of edge-count variables is in general hard with the RGFDS. This implies that slow numerical methods based on simulations must be used [13, 15, 14].

A less constrained random graph model than the RGFDS is the Random Graph with Given Expected Degrees (RGGED) [7, 8]. While an individual realization of the RGGED might not preserve a vertex degree, it is preserved on average over all realizations. The key feature of the RGGED is that edges are treated as independent random variables, enabling the derivation of analytical expressions for the distributions of edge-count variables [18].

In the RGGED model, edges are treated as independent Bernoulli random variables: an edge $v_i v_j$ exists with probability p_{ij} (Bernoulli parameter) and does not exist with probability $1 - p_{ij}$. When the graph is large and sparse on average over its vertices ($\langle k \rangle^2 \ll 2M$), the probability p_{ij} of observing an edge between vertices v_i and v_j is given by [18]:

$$p_{ij} \simeq 1 - \exp(-k_i k_j / (2M)). \quad (1)$$

Useful upper bounds for p_{ij} are

$$p_{ij} \leq \min\left(1, \frac{k_i k_j}{2M}\right) \leq \frac{k_i k_j}{2M}. \quad (2)$$

2.1 Within and between variables

The `EdgeCount` package considers two types of edge-count variables:

- ”within” for the number z of edges observed in a given set S of vertices

- "between" for the number z of edges observed between two sets S_1 and S_2 of vertices

In this framework, the sets and the degrees of their constituent vertices are fixed parameters. Observed values z are thus conditional on these degrees. A corresponding random variable Z in the RGGED is associated to an observed value z ; its distribution enables assessment of the significance of z .

2.2 Edge-count distribution

Edge-count random variables in the RGGED model of a large and sparse graph have an approximate Poisson distribution. This is because they are the sum of independent Bernoulli random variables with small parameters $p_{ij} \ll 1$, even when these parameters have different values [12, 11, 18].

Call Z an edge-count random variable defined by the sum of m potential edges, each edge corresponding to a pair $v_i v_j$ of vertices:

$$Z = \sum_{u=1}^m B_u \quad \text{with} \quad B_u \sim \text{Bernoulli}(p_u = p_{ij}). \quad (3)$$

Then, we have $Z \sim \text{Poisson}(\lambda, m)$ as defined by

$$\Pr(Z = z) = \frac{e^{-\lambda} \lambda^z}{\alpha z!} \quad \text{with} \quad \lambda = \sum_{u=1}^m p_u \quad \text{and} \quad \alpha = e^{-\lambda} \sum_{x=0}^m \frac{\lambda^x}{x!}. \quad (4)$$

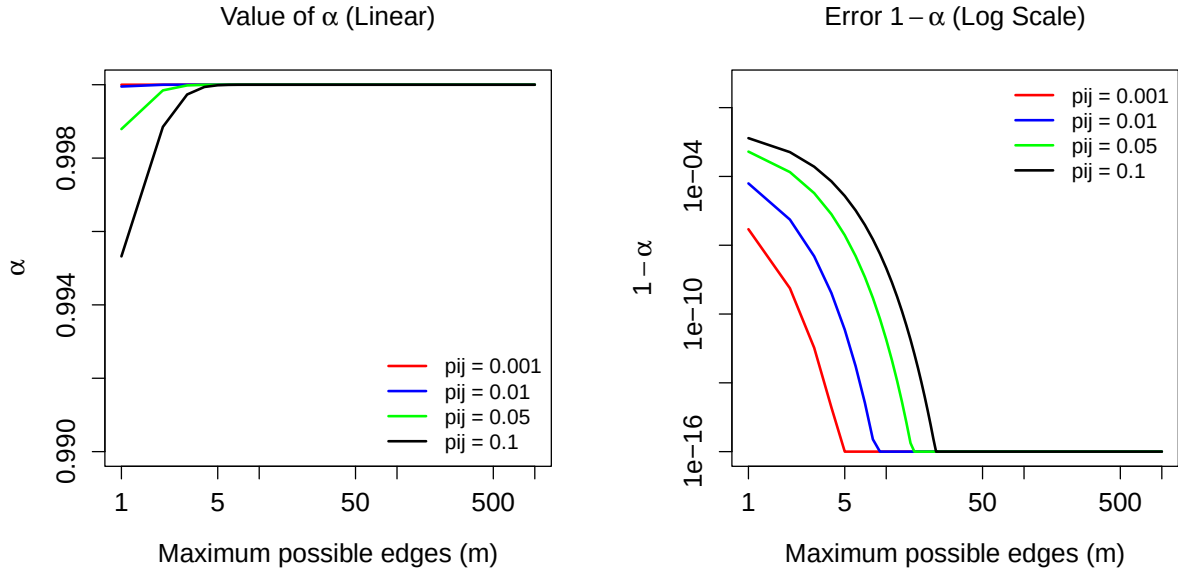
The normalization factor α is utilized because an observed number z of edges is bounded by a maximum number m of edges. This said, we have $\alpha \simeq 1$ when $p_{ij} \ll 1$. This is numerically illustrated next.

```
# function to calculate alpha and its complement
get_alpha_stats <- function(m, pij_avg) {
  lambda <- m * pij_avg
  alpha <- stats::ppois(m, lambda)
  alpha_comp <- stats::ppois(m, lambda, lower.tail = FALSE)
  return(c(alpha, alpha_comp))
}
# parameters
pij_values <- c(0.001, 0.01, 0.05, 0.1)
m_values <- c(1:29, seq(30, 1000, by = 10))
colors <- c("red", "blue", "green", "black")
# plots
par(mfrow=c(1, 2), mar=c(5, 5, 4, 2) + 0.1, cex.lab=1.25,
    cex.axis=1.25, cex.main=1.25)
plot(1, type="n", xlab="Maximum possible edges (m)",
     ylab=expression(alpha),
     xlim=c(1, 1000), ylim=c(0.99, 1.00),
     log="x",
     main=expression(paste("Value of ", alpha, " (Linear)")))
for(i in seq_along(pij_values)) {
  lambdas <- m_values * pij_values[i]
  y_vals <- ppois(m_values, lambdas)
  lines(m_values, y_vals, col=colors[i], lwd=2)
}
legend("bottomright", legend=paste("pij =", pij_values),
      col=colors, lwd=2, bty="n", cex=1)
epsilon <- 1e-16
plot(1, type="n", xlab="Maximum possible edges (m)",
     ylab=expression(1 - alpha),
     xlim=c(1, 1000), ylim=c(epsilon, 1),
```

```

log="xy",
main=expression(paste("Error ", 1 - alpha, " (Log Scale)"))
for(i in seq_along(pij_values)) {
  lambdas <- m_values * pij_values[i]
  y_vals <- ppois(m_values, lambdas, lower.tail = FALSE)
  y_vals <- pmax(y_vals, epsilon)
  lines(m_values, y_vals, col=colors[i], lwd=2)
}
legend("topright", legend=paste("pij =", pij_values),
      col=colors, lwd=2, bty="n", cex=1)

```



For a within variable associated with a set S we have

$$m = |S|(|S| - 1)/2 \quad \text{and} \quad \lambda = \sum_{i < j, i, j \in S} p_{ij}. \quad (5)$$

For a between variable associated with two sets S_1 and S_2 we have

$$m = |S_1||S_2| \quad \text{and} \quad \lambda = \sum_{i \in S_1, j \in S_2} p_{ij}. \quad (6)$$

Edge-count p-values are given by

$$\Pr(Z \geq z) = \sum_{x=z}^m \Pr(Z = x). \quad (7)$$

P-values are calculated with R function `ppois()`.

```

alpha <- stats::ppois(m, lambda, lower.tail = TRUE)
p_value <- (stats::ppois(z - 1, lambda, lower.tail = FALSE) -
  stats::ppois(m, lambda, lower.tail = FALSE)) / alpha

```

Note that the Poisson distribution enables quantification of the variability on vertex degree modeled by the RGGED. In the model, the degree of a vertex v_i with initial degree k_i is a random variable K_i defined by

the edge count between this vertex and all the others:

$$\lambda_i = \frac{k_i}{2M} \sum_{j \neq i} k_j \simeq k_i. \quad (8)$$

That is, K_i has mean k_i and standard deviation $\sqrt{k_i}$. The introduced relative variability on k_i is reasonable: $1/\sqrt{k_i}$.

2.3 Edge-count effect size

As explained earlier, the RGGED is not always directly utilized as a null model to answer a question. Instead, it can be used as an intermediate step that enables fair and fast comparison of diverse observed edge counts. Such a comparison can be between edge-count variables defined by sets of different sizes. In this case, an effect size is the most appropriate quantity for comparison.

Because edge-count random variables have Poisson distribution, the utilized effect size is

$$\log_2 \text{Anscombe ratio: } lAr(z, \lambda) = \frac{1}{2} \log_2 \frac{z + 3/8}{\lambda + 3/8}. \quad (9)$$

The addition of $3/8$ stems from the work of Anscombe [1], who showed that $\sqrt{X + 3/8}$ is the optimal variance-stabilizing transformation for a Poisson random variable X .

3 Fast methods for sparse interaction networks

All the code for representation and analysis of undirected unipartite graphs (interaction networks) is contained in S4 classes `ECGraph` and `ECProb`.

3.1 Sparseness condition

A condition for sparseness in the `EdgeCount` framework is

$$\max_{i,j} \frac{k_i k_j}{2M} < 1. \quad (10)$$

This sparseness condition can be tested with method `summarize_suitability_fast()` of class `ECProb`. The method returns a list containing the proportion `pj_over_1`, among all possible pairs v_i, v_j of vertices, of pairs such that $(k_i k_j)/(2M) \geq 1$. An ideally sparse graph yields `pj_over_1` = 0.

Now, approximating p_{ij} values by $k_i k_j/(2M)$ is a conservative approach in the RGGED. Namely, using this approximation

- overestimates p_{ij} values,
- overestimates lambda values,
- overestimates edge-count p-values,
- underestimates effect sizes.

Therefore, utilizing this approximation by default is a reasonable approach. The approximation is called "fast approximation" because it enables faster calculation of edge-count statistics. The speedup is obtained in the computation of lambda parameters.

3.2 Within-set connectedness

Consider a vertex set S of size $|S| = n$. With the fast approximation the λ parameter is

$$\lambda_{in} = \sum_{i < j, i \in S} \frac{k_i k_j}{2M}. \quad (11)$$

A naive implementation would have time complexity $\mathcal{O}(n^2)$. Recall the expression for the square of a sum

$$\left(\sum_{i \in S} k_i \right)^2 = \sum_{i \in S} k_i^2 + 2 \sum_{i < j} k_i k_j. \quad (12)$$

Rearranging the cross terms

$$\sum_{i < j} k_i k_j = \frac{1}{2} \left[\left(\sum_{i \in S} k_i \right)^2 - \sum_{i \in S} k_i^2 \right], \quad (13)$$

and substituting in the expression for lambda yields

$$\lambda_{in} = \frac{1}{4M} \left[\left(\sum_{i \in S} k_i \right)^2 - \sum_{i \in S} k_i^2 \right]. \quad (14)$$

With this expression the computation of λ_{in} has now linear time complexity: $\mathcal{O}(n)$. Method `edge_count_statistics()` of class `ECProb` when used with argument `lambda_method = "fast"` implements this linear computation. Method `calculate_in_stats_fast_vectorized()` is an implementation of "within" edge count statistics for multiple sets provided as input. This method is fully vectorized and uses the `data.table` package [2]. Here is a numerical illustration that the implementation achieves optimal, linear here, time complexity.¹ The illustration is based on the interaction network `sample_ecg` contained in the `EdgeCount` package and derived from the STRING database [23, 20].

```
library(data.table)
library(EdgeCount)
data("sample_ecg")
ecprob <- ECProb(sample_ecg)
n_sim <- 5
n_collections <- 10
collection_size_universe <- seq(10, 1000, length.out = n_collections)
set_size_universe <- 10:100
set.seed(1)

get_collection <- function(target_n_sets){
  n_sets <- round(target_n_sets)
  sets <- lapply(1:n_sets, function(x){
    set_size <- sample(set_size_universe, 1)
    sample(ecprob@names, set_size)
  })
  set_membership_dt <- data.table(
    set_id = rep(seq_along(sets), lengths(sets)),
    element = unlist(sets)
  )
  return(set_membership_dt)
}
```

¹Such numerical results are always dependent on the utilized CPU. A faster machine might require larger simulated sets to clearly evidence linearity.

```

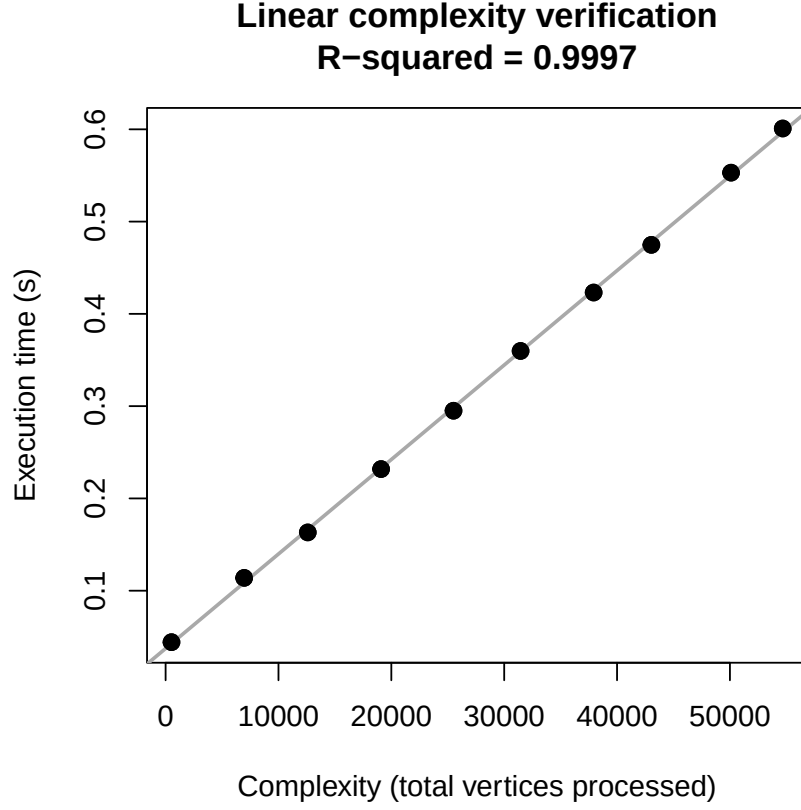
collections <- lapply(collection_size_universe, get_collection)
results_list <- lapply(collections, function(set_membership_dt){
  sets_dt <- data.table(set_id = unique(set_membership_dt$set_id))
  times <- numeric(n_sim)
  res <- NULL
  for(k in 1:n_sim){
    start_time <- Sys.time()
    capture.output(
      res <- calculate_in_stats_fast_vectorized(ecprob, sets_dt, set_membership_dt)
    )
    end_time <- Sys.time()
    times[k] <- as.numeric(end_time - start_time, units = "secs")
  }
  complexity <- sum(res$set_size, na.rm = TRUE)
  data.table(complexity = complexity, time = min(times))
})

results_dt <- rbindlist(results_list)

lm_model <- lm(time ~ complexity, data = results_dt)
r_sq <- summary(lm_model)$r.squared

plot(results_dt$complexity, results_dt$time, pch=19, cex=1.2,
      xlab = "Complexity (total vertices processed)",
      ylab = "Execution time (s)",
      main = paste("Linear complexity verification\nR-squared =", round(r_sq, 4)))
abline(lm_model, col = "darkgrey", lwd=2)
points(results_dt$complexity, results_dt$time, pch=19, cex=1.2)

```



The execution time scales linearly (R-squared value close to 1) with the optimal time complexity, which is the sum of the set sizes in each set collection used as input.

The fast approximation of p_{ij} also enables optimal computation of "between" connectedness statistics.

3.3 Between-sets connectedness

Consider two disjoint vertex sets S_1 and S_2 with sizes n_1 and n_2 . With the fast approximation, the λ parameter is

$$\lambda_{btw} = \sum_{i \in S_1, j \in S_2} \frac{k_i k_j}{2M}. \quad (15)$$

A naive implementation summing over all pairs would have time complexity $\mathcal{O}(n_1 n_2)$. However, the expression can be factored as the product of two independent sums:

$$\lambda_{btw} = \frac{1}{2M} \left(\sum_{i \in S_1} k_i \right) \left(\sum_{j \in S_2} k_j \right). \quad (16)$$

This factorization reduces the calculation to simply summing the degrees within each set and multiplying the results. Consequently, the computation of λ_{btw} has linear time complexity: $\mathcal{O}(n_1 + n_2)$.

Fast calculation of 'between' edge-count statistics for a single pair of sets is implemented in method `edge_count_statistics()` of class `ECProb` by setting both arguments `set1` and `set2` to non-null vectors and by using `lambda_method = "fast"`.

For multiple pairs of sets, the appropriate method is `calculate_between_stats_fast_vectorized()`. It implements the fast approximation and is fully vectorized in parts by using the `data.table` package [2]. The

following simulation confirms that it achieves optimal, linear here, time complexity.²

```
library(data.table)
library(EdgeCount)
data("sample_ecg")
ecprob <- ECPProb(sample_ecg)

n_sim <- 10
n_collections <- 10
n_pairs_universe <- seq(500, 5000, length.out = n_collections)
set_size_universe <- 10:50
set.seed(123)

get_between_collection <- function(n_pairs){

  n_pairs <- round(n_pairs)
  total_sets <- 2 * n_pairs

  sets <- lapply(1:total_sets, function(x) {
    sample(ecprob@names, sample(set_size_universe, 1))
  })

  set_membership_dt <- data.table(
    set_id = rep(seq_along(sets), lengths(sets)),
    element = unlist(sets)
  )

  pairs_dt <- data.table(
    set1 = seq(1, total_sets, by = 2),
    set2 = seq(2, total_sets, by = 2)
  )

  return(list(pairs = pairs_dt, membership = set_membership_dt))
}

collections_btw <- lapply(n_pairs_universe, get_between_collection)

results_list_btw <- lapply(collections_btw, function(input_data){

  pairs_dt <- input_data$pairs
  membership_dt <- input_data$membership

  times <- numeric(n_sim)
  res <- NULL

  for(k in 1:n_sim){
    start_time <- Sys.time()
    temp_res <- NULL
    capture.output(
      temp_res <- calculate_between_stats_fast_vectorized(ecprob, pairs_dt, membership_dt)
    )
    res <- temp_res
  }
}
```

²Such numerical results are always dependent on the utilized CPU. A faster machine might require larger simulated sets to clearly evidence linearity.

```

end_time <- Sys.time()
times[k] <- as.numeric(end_time - start_time, units = "secs")
}

complexity <- sum(res$size1, na.rm = TRUE) + sum(res$size2, na.rm = TRUE)

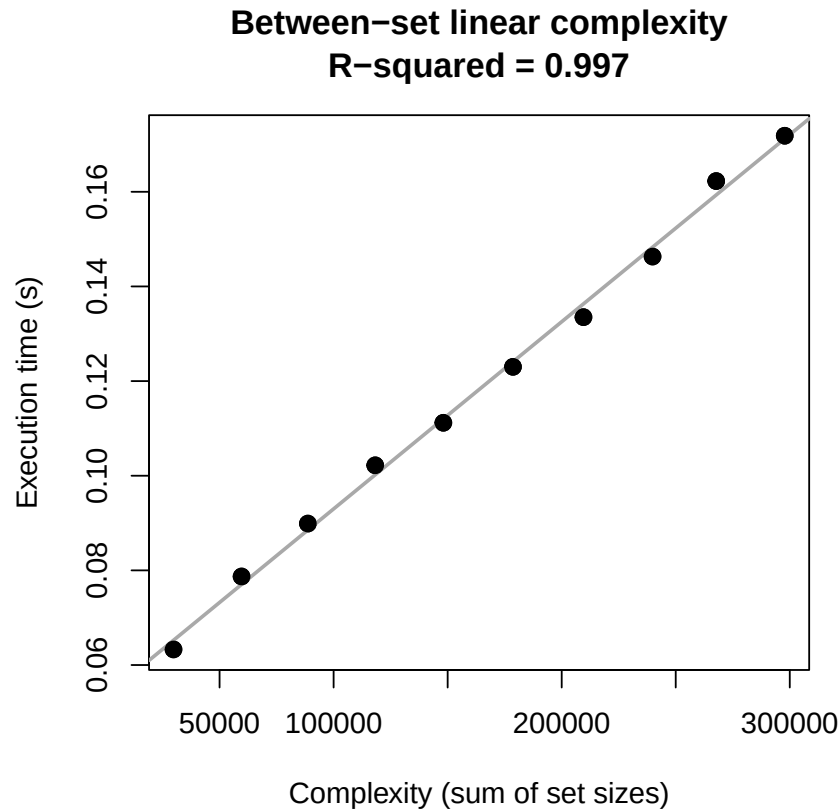
data.table(complexity = complexity, time = min(times))
})

results_dt_btw <- rbindlist(results_list_btw)

lm_model_btw <- lm(time ~ complexity, data = results_dt_btw)
r_sq_btw <- summary(lm_model_btw)$r.squared

plot(results_dt_btw$complexity, results_dt_btw$time, pch=19, cex=1.2,
      xlab = "Complexity (sum of set sizes)",
      ylab = "Execution time (s)",
      main = paste("Between-set linear complexity\nR-squared =", round(r_sq_btw, 4)))
abline(lm_model_btw, col = "darkgrey", lwd=2)
points(results_dt_btw$complexity, results_dt_btw$time, pch=19, cex=1.2)

```



4 Fast methods for sparse bipartite networks

Class `ECTermScoring` complements the functionality of `ECGraph` and `ECProb` to the case of undirected bipartite graphs. These represent here membership of elements within groups, called terms in the package. The

package comes with a sample bipartite network `sample_ects` that maps a trimmed version of GO categories to their human gene elements [6, 24, 9].

4.1 Sparseness condition

Call $\mathcal{T}$ the set of vertices that represent terms and $\mathcal{E}$ the set of elements. The sparseness condition is:

$$\max_{i \in \mathcal{T}, j \in \mathcal{E}} \frac{k_i k_j}{2M} < 1. \quad (17)$$

This is the sparseness condition already presented for interaction networks, but adapted to edges existing only between terms and elements. Method `summarize_suitability_fast()` of class `ECTermScoring` is used to test this condition. The method returns a list containing the proportion `pj_over_1`, among all possible pairs v_i, v_j , of values $(k_i k_j)/(2M) \geq 1$. An ideally sparse graph yields `pj_over_1` = 0.

But as mentioned for interaction networks, approximating p_{ij} values with $k_i k_j/(2M)$ is a conservative approach in the RGGED. Namely, using this approximation

- overestimates p_{ij} values,
- overestimates lambda values,
- overestimates edge-count p-values,
- underestimates effect sizes.

Therefore, utilizing this approximation by default is a reasonable approach.

4.2 Fast approximation

The fast approximation implemented in `ECTermScoring` is analogous to that used in `ECProb` for between-sets connectedness. Consider a set S of elements and its term neighborhood T_S (terms connected to at least one element in S).

The expected number of edges λ in this subgraph is given by the double summation:

$$\lambda = \sum_{t \in T_S} \sum_{e \in S} p_{et}. \quad (18)$$

A naive calculation would have multiplicative time complexity $\mathcal{O}(|S| \cdot |T_S|)$. However, utilizing the fast approximation $p_{et} = k_e k_t/(2M)$, the expression factors into the product of two sums:

$$\lambda = \frac{1}{2M} \left(\sum_{t \in T_S} k_t \right) \left(\sum_{e \in S} k_e \right). \quad (19)$$

Consequently, the time complexity for scoring all terms in the neighborhood becomes additive: $\mathcal{O}(|S| + |T_S|)$. More generally, the average time complexity for a random set S of elements scales as:

$$\mathcal{O}(|S| (1 + \langle k_e \rangle)), \quad (20)$$

where $\langle k_e \rangle$ is the average element degree across the graph.

4.3 Scoring terms against element sets

The primary analysis on bipartite graphs is identifying which terms are significantly connected to a given set of elements.

The basic method `terms_ecset_statistics()` calculates connectedness statistics for a single set of elements against all its connected terms. These RGGED-based statistics can then be utilized to rank terms for their connection strength to the set.

This said, analyses often require scoring multiple sets simultaneously. The method optimized for this purpose is `terms_ecset_statistics_vectorized()`. By utilizing the fast approximation derived in the previous section and `data.table` operations [2], it achieves the desired linear time complexity. The following simulation verifies this linear scaling.³ As explained before, linear complexity is the sum of the number of input elements and the number of connected terms (neighborhood size).

```
library(data.table)
library(EdgeCount)
data("sample_ects")

# simulation
n_sim <- 5
n_points <- 10
n_sets_universe <- seq(50, 1000, length.out = n_points)
set.seed(1)
p_values <- numeric()

results_list <- lapply(n_sets_universe, function(n_s){
  n_s <- round(n_s)
  set_sizes <- sample(10:50, n_s, replace = TRUE)
  elements <- unlist(lapply(set_sizes, function(sz) {
    sample(sample_ects@elements, sz)
  })))

  input_sets_dt <- data.table(
    set_id = rep(paste0("S", 1:n_s), times = set_sizes),
    element = elements
  )

  times <- numeric(n_sim)
  res <- NULL

  for(k in 1:n_sim){
    start_time <- Sys.time()
    capture.output(
      res <- terms_ecset_statistics_vectorized(sample_ects, input_sets_dt)
    )
    end_time <- Sys.time()
    times[k] <- as.numeric(end_time - start_time, units = "secs")

    if (k == 1){
      current_p <- rbindlist(res)$p_value
      p_values <- c(p_values, current_p)
    }
  }

  neighborhood_size <- sum(vapply(res, nrow, integer(1)))
  complexity <- nrow(input_sets_dt) + neighborhood_size

  data.table(complexity = complexity, time = min(times))
})
```

³Such numerical results are always dependent on the utilized CPU . A faster machine might require larger simulated sets to clearly evidence linearity.

```

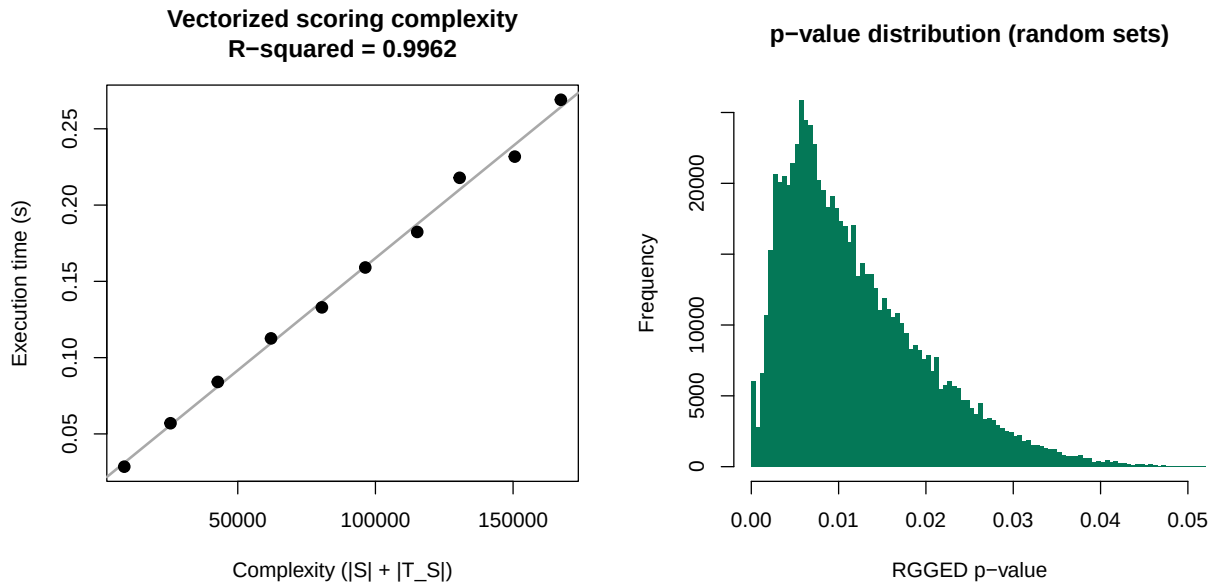
results_dt <- rbindlist(results_list)
lm_model <- lm(time ~ complexity, data = results_dt)
r_sq <- summary(lm_model)$r.squared

# plots
par(mfrow=c(1, 2), mar=c(5, 4, 4, 2) + 0.1)

plot(results_dt$complexity, results_dt$time, pch=19,
      xlab = "Complexity (|S| + |T_S|)",
      ylab = "Execution time (s)",
      main = paste("Vectorized scoring complexity\nR-squared =", round(r_sq, 4)))
abline(lm_model, col="darkgrey", lwd=2)
points(results_dt$complexity, results_dt$time, pch=19, cex=1.2)

hist(p_values, breaks = 100, col = "#047857", border = NA,
      main = "p-value distribution (random sets)",
      xlab = "RGGED p-value",
      xlim = c(0, 0.05))

```



```

summary(p_values)

```

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	1.220e-07	5.771e-03	1.005e-02	1.202e-02	1.659e-02	6.595e-02

The left panel of the figure shows that the implementation of the method achieves the desired time complexity of $\mathcal{O}(|S| + |T_S|)$.

4.4 Empirical false discovery rate

The right panel of the last figure shows the distribution of RGGED p-values obtained with the simulation of random element sets. It is clearly non-uniform with a median value close to 0.01. This is because methods used to score sets against terms return only statistics for connected terms (edge count $z \geq 1$). This implies that the RGGED cannot be directly used here as a model for the random experiment of selecting a random set. Further modeling is required. Moreover, many sets will return multiple terms, implying that correction for multiple testing is required. Because the distribution of RGGED p-values is non-uniform, standard multiple

testing corrections (like Benjamini-Hochberg)[5] are not applicable.

Consequently, the package implements an Empirical FDR approach in `terms_ecset_statistics_fdr()`. This method:

1. Generates a null distribution of effect sizes by simulating random element sets of the same size as the input (reduced to the element universe in the term-element network).
2. Calculates a Normalized Enrichment Score (NES) for each term.
3. Estimates the False Discovery Rate (q-value) by comparing the observed NES to its null distribution.

Since this approach requires repeated simulations, the implementation must scale efficiently. The following simulation confirms that the runtime of `terms_ecset_statistics_fdr()` scales linearly with the requested number of permutations.⁴

```
set.seed(1)
test_set <- sample(sample_ects@elements, 30)
n_perms_universe <- seq(100, 2000, length.out = 10)
n_sim <- 5

results_fdr <- lapply(n_perms_universe, function(n_p){
  n_p <- round(n_p)
  times <- numeric(n_sim)

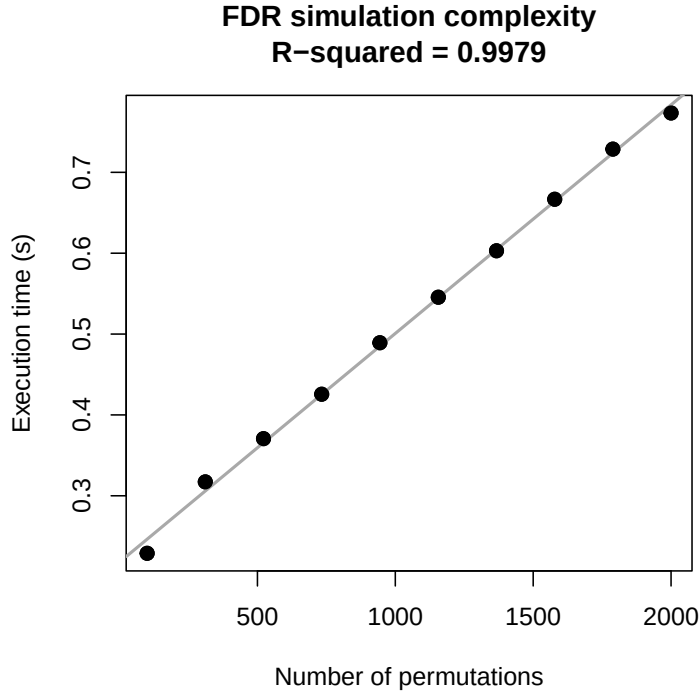
  for(k in 1:n_sim){
    start_time <- Sys.time()
    capture.output(
      res <- terms_ecset_statistics_fdr(sample_ects,
                                       test_set,
                                       n_permutations = n_p,
                                       seed = NULL)
    )
    end_time <- Sys.time()
    times[k] <- as.numeric(end_time - start_time, units = "secs")
  }

  data.table(permutations = n_p, time = min(times))
})

results_fdr_dt <- rbindlist(results_fdr)
lm_fdr <- lm(time ~ permutations, data = results_fdr_dt)
r_sq_fdr <- summary(lm_fdr)$r.squared

plot(results_fdr_dt$permutations, results_fdr_dt$time, pch=19, cex=1.2,
      xlab = "Number of permutations",
      ylab = "Execution time (s)",
      main = paste("FDR simulation complexity\nR-squared =", round(r_sq_fdr, 4)))
abline(lm_fdr, col="darkgrey", lwd=2)
points(results_fdr_dt$permutations, results_fdr_dt$time, pch=19, cex=1.2)
```

⁴Such numerical results are always dependent on the utilized CPU. A faster machine might require larger simulated sets to clearly evidence linearity.



5 Vertex Set Enrichment Analysis (VSEA)

While the analytical methods described in previous sections enable fast computation of edge-count statistics for specific sets, users often wish to analyze a ranked list of elements to detect enrichment patterns at the top of the list. Method `run_vsea()` is analogous to Gene Set Enrichment Analysis (GSEA) [22], but it is applied to graph topology. In particular, it takes into account the degrees of elements, therefore penalizing scores when ubiquitous elements, those linked to many terms, are present at the top of the ranked list.

The `run_vsea()` method calculates Normalized Enrichment Scores (NES) and Empirical False Discovery Rates (FDR) using permutation-based simulations.

5.1 Methodology and implementation

The analysis proceeds in three steps:

- **Conditioning:** The analysis universe is restricted to the intersection of the graph universe and the provided list of ranked elements.
- **Scoring:** The list is traversed to calculate the Log Anscombe Ratio (lAr) for every term at every rank. The maximum score for each term is retained.
- **Significance:** A null distribution is generated by permuting the element ranks N times, allowing for the calculation of NES and FDR.

The implementation relies on parallelization and low-level vectorization:

- **Pre-computation:** Cumulative degrees for all permutations are pre-calculated efficiently using `matrixStats` [4].
- **Fast randomization:** Random permutations are generated using the `dqrng` package [21].
- **Radix sorting:** Inside each simulation, the edges induced by the random permutation are identified and sorted by term using `order(..., method="radix")`. This allows for the rapid calculation of running

scores using Run-Length Encoding (`rle`), avoiding slow R loops.

- Parallel processing: Simulations are distributed across CPU cores using `parallel::mclapply`[19].

For a dataset with N_e elements with ranking scores and a set of terms $\mathcal{T}$ where term t has degree k_t , the complexity scales as:

$$\mathcal{O}\left(N_e \log(N_e) + \sum_{t \in \mathcal{T}} k_t \log(k_t)\right). \quad (21)$$

5.2 Scaling with data size

The time complexity is composed of two distinct parts:

- ranking elements: Scales as $\mathcal{O}(N_e \log(N_e))$.
- sorting Edges within Terms: Scales as $\mathcal{O}(\sum k_t \log(k_t))$.

A simulation varying the universe size is performed. Since the two operations incur different computational costs, a multiple linear regression model is fitted to the observed times.⁵

Note: The graph objects are pre-reduced and the input ranks exactly match the graph universe. This bypasses the conditional reduction step inside `run_vsea`, so as to measure the pure scoring performance.

```
data("sample_ects")
set.seed(1)

# terms with degree >= 4
all_terms <- sample_ects@terms
term_dt <- data.table(term = all_terms,
                      d = unlist(sample_ects@ecprob@degrees[all_terms]))
valid_terms <- term_dt[d >= 4, term]
full_ects <- reduce_universe_by_terms(sample_ects, valid_terms)

# simulation parameters
term_counts <- round(seq(200, 3000, length.out = 10))
n_sim <- 5
fixed_perms <- 50

results_list <- lapply(term_counts, function(n_t){

  curr_terms <- sample(full_ects@terms, n_t)
  ects_sub <- reduce_universe_by_terms(full_ects, curr_terms)
  ne <- length(ects_sub@elements)
  kt <- lengths(ects_sub@ecprob@adj[ects_sub@terms])

  ne_log_ne <- ne * log(ne)
  sum_kt_log_kt <- sum(kt * log(kt + 1e-9))

  ranks <- setNames(sample(seq_len(ne)), ects_sub@elements)
  times <- numeric(n_sim)
  for(k in 1:n_sim){
    gc(verbose = FALSE)
    start_time <- Sys.time()
    capture.output(
      run_vsea(ects_sub, ranks, n_permutations = fixed_perms,
```

⁵Such numerical results are always dependent on the utilized CPU. A faster machine might require larger simulated sets to clearly evidence linearity.


```

        n_cores = 1, seed = NULL)
    )
    end_time <- Sys.time()
    times[k] <- as.numeric(end_time - start_time, units = "secs")
  }

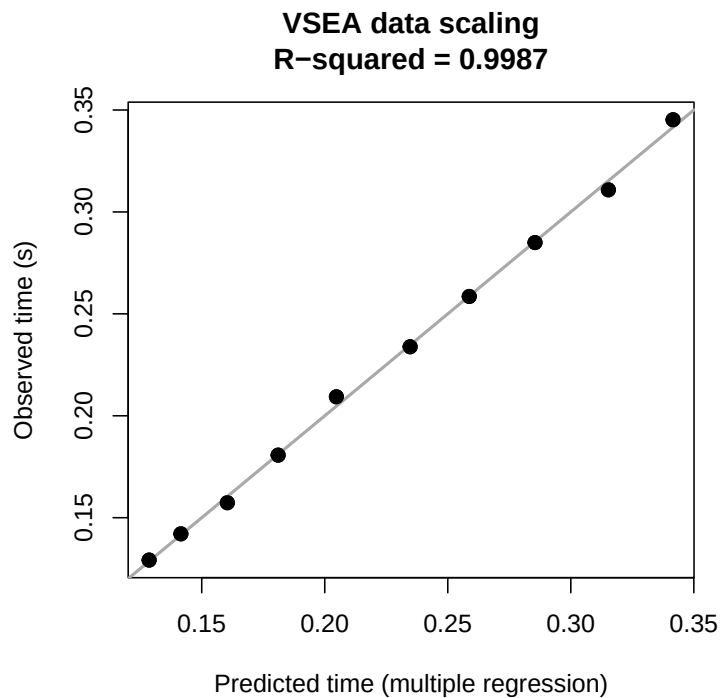
  data.table(ne_log_ne = ne_log_ne,
             sum_kt_log_kt = sum_kt_log_kt,
             time = min(times))
})

results_dt <- rbindlist(results_list)

# multiple linear model
lm_model <- lm(time ~ ne_log_ne + sum_kt_log_kt, data = results_dt)
r_sq <- summary(lm_model)$r.squared

# plots predicted (fitted) vs observed time
plot(predict(lm_model), results_dt$time, pch=19, cex=1.2,
     xlab = "Predicted time (multiple regression)",
     ylab = "Observed time (s)",
     main = paste("VSEA data scaling\nR-squared =", round(r_sq, 4)))
abline(0, 1, col="darkgrey", lwd=2)
points(predict(lm_model), results_dt$time, pch=19, cex=1.2)

```



5.3 Scaling with number of simulations

The last step is to verify that the implementation scales linearly with the requested number of permutations. The simulation below confirms the linear scaling.⁶

```
data("sample_ects")
ranks <- setNames(sample(seq_along(sample_ects@elements)),
                  sample_ects@elements)

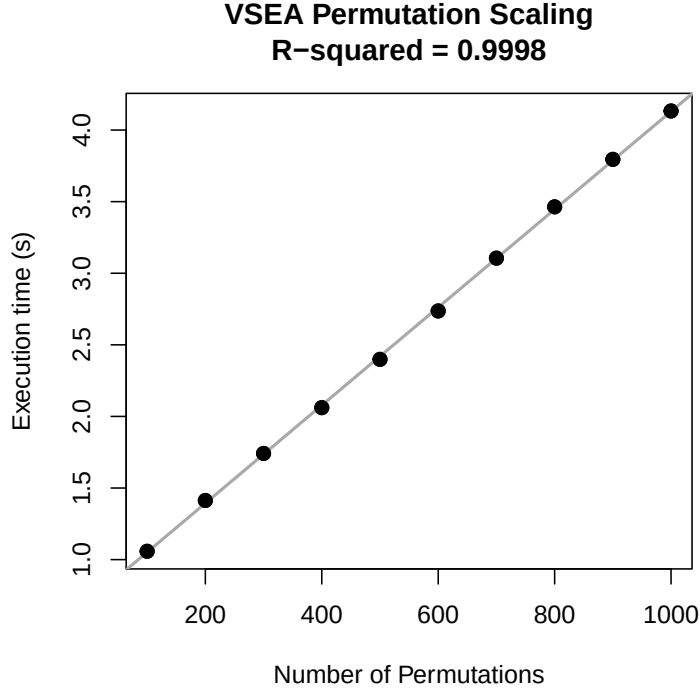
perm_counts <- seq(100,1000, length.out = 10)
n_sim <- 5

results_perm <- lapply(perm_counts, function(n_p){
  times <- numeric(n_sim)
  for(k in 1:n_sim){
    start_time <- Sys.time()
    capture.output(
      run_vsea(sample_ects, ranks, n_permutations = n_p,
               n_cores = 1, seed = NULL)
    )
    end_time <- Sys.time()
    times[k] <- as.numeric(end_time - start_time, units = "secs")
  }
  data.table(permutations = n_p, time = min(times))
})

results_perm_dt <- rbindlist(results_perm)
lm_perm <- lm(time ~ permutations, data = results_perm_dt)
r_sq_perm <- summary(lm_perm)$r.squared

plot(results_perm_dt$permutations, results_perm_dt$time,
     pch=19, cex=1.2,
     xlab = "Number of Permutations",
     ylab = "Execution time (s)",
     main = paste("VSEA Permutation Scaling\nR-squared =", round(r_sq_perm, 4)))
abline(lm_perm, col="darkgrey", lwd=2)
points(results_perm_dt$permutations, results_perm_dt$time, pch=19, cex=1.2)
```

⁶Such numerical results are always dependent on the utilized CPU. A faster machine might require larger simulated sets to clearly evidence linearity.



5.4 Conclusion

The time complexity of VSEA as implemented in method `run_vsea` is

$$\mathcal{O} \left(N_s \left(N_e \log(N_e) + \sum_t k_t \log(k_t) \right) \right), \quad (22)$$

where N_s is the number of simulations for FDR estimation, N_e the number of elements connected to at least one term in the bipartite network and k_t the number of elements, included in the graph and the input rank list, a term t connects to.

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